

COMPLETE STUDY GUIDE · VOL. 1

Mathematics Fundamentals

From School Basics to Pre-College Mastery · 10 Modules · 80+
Solved Problems · Formula Sheets

Number Systems

Fractions & Decimals

Algebra

Exponents & Radicals

Factoring

Quadratics

Geometry

Trigonometry Intro

Statistics

Probability

Video Links Included · Learn by Doing

Table of Contents

1

Number Systems & BODMAS

Integers, Rationals, Order of Operations

2

Fractions, Decimals & Percentages

Operations, Conversions, Percentage Change

3

Ratios & Proportions

Direct, Inverse, Word Problems

4

Linear Algebra Basics

Equations, Inequalities, Word Problems

5

Simultaneous Equations

Substitution, Elimination, Applications

6

Exponents & Radicals

Laws of Indices, Surds, Simplification

7

Polynomials & Factoring

Operations, GCF, Trinomials, Identities

8

Quadratic Equations

Factoring, Formula, Discriminant, Graphs

9

Geometry & Mensuration

Triangles, Circles, 3D Shapes, Pythagoras

10

Statistics & Probability

Mean, Median, Variance, Basic Probability

Number Systems & Order of Operations

Before tackling algebra or calculus, you must master how numbers are classified and how to evaluate expressions correctly. These are the absolute foundations every other concept builds upon.

Types of Numbers

Numbers are organised in sets, each a superset of the one before it: Natural $\subset$ Whole $\subset$ Integer $\subset$ Rational $\subset$ Real. Understanding which set a number belongs to tells you what operations are valid.

► Watch: Algebra Introduction – Video-Tutor.net

NUMBER SETS

Natural ($\mathbb{N}$)

1, 2, 3, 4, ...

Whole ($\mathbb{W}$)

0, 1, 2, 3, ...

Integers ($\mathbb{Z}$)

..., -2, -1, 0, 1, 2, ...

Rational ($\mathbb{Q}$)

$\frac{p}{q}$, $q \neq 0$, $p, q \in \mathbb{Z}$

Irrational

$\sqrt{2}$, π , e (non-terminating, non-repeating)

Real ($\mathbb{R}$)

Rational $\cup$ Irrational

1 Classify each number into ALL applicable sets

Classify: -7 , $\frac{3}{4}$, $\sqrt{5}$, 0 , $\sqrt{9}$

STEP-BY-STEP SOLUTION

- 1 -7 : Negative whole number $\rightarrow$ Integer, Rational, Real (not Natural, not Whole)
- 2 $\frac{3}{4}$: Fraction of two integers, $q \neq 0 \rightarrow$ Rational, Real only

- 3 $\sqrt{5} \approx 2.236 \dots$: Non-terminating, non-repeating $\rightarrow$ Irrational, Real only
- 4 0: Zero $\rightarrow$ Whole, Integer, Rational ($0 = \frac{0}{1}$), Real
- 5 $\sqrt{9} = 3$: Perfect square, equals 3 $\rightarrow$ Natural, Whole, Integer, Rational, Real

Answer: $\sqrt{9}$ belongs to ALL five sets; $\sqrt{5}$ belongs to Irrational & Real only

2 True or False reasoning

State True or False and justify: (a) Every integer is rational. (b) Every rational is an integer. (c) π is real but not rational.

SOLUTION

- a **True.** Any integer n can be written as $\frac{n}{1}$, which is rational.
- b **False.** $\frac{1}{2}$ is rational but not an integer (no whole-number form).
- c **True.** $\pi \approx 3.14159 \dots$ never repeats or terminates $\rightarrow$ Irrational $\subset$ Real.

Answer: True, False, True

Order of Operations (BODMAS / PEMDAS)

When an expression has multiple operations, you must follow a strict priority order to get the correct answer every time.

BODMAS RULE

B – Brackets

Innermost first: () then [] then { }

O – Orders

Powers and roots: x^n , $\sqrt{x}$

D/M – $\div$ $\times$

Left to right: Division and Multiplication

A/S – $+$ $-$

Left to right: Addition and Subtraction

1 Evaluate using BODMAS

Calculate: $3 + 6 \times (5 + 4) \div 3 - 7$

STEP-BY-STEP SOLUTION

- 1 Bracket first: $5 + 4 = 9 \rightarrow$ expression becomes $3 + 6 \times 9 \div 3 - 7$

- 2 Multiply: $6 \times 9 = 54 \rightarrow 3 + 54 \div 3 - 7$
- 3 Divide: $54 \div 3 = 18 \rightarrow 3 + 18 - 7$
- 4 Add/Subtract left to right: $3 + 18 = 21$, then $21 - 7 = 14$

Answer: 14

2 Nested brackets & exponents

Evaluate: $2^3 + [18 - (6 \times 2)] \div 2$

STEP-BY-STEP SOLUTION

- 1 Innermost bracket: $6 \times 2 = 12 \rightarrow 2^3 + [18 - 12] \div 2$
- 2 Square bracket: $18 - 12 = 6 \rightarrow 2^3 + 6 \div 2$
- 3 Order (exponent): $2^3 = 8 \rightarrow 8 + 6 \div 2$
- 4 Divide: $6 \div 2 = 3 \rightarrow 8 + 3$
- 5 Add: $8 + 3 = 11$

Answer: 11

Module 1 — Formula & Rule Summary

Number Hierarchy

$$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

Rational Form

$$\frac{p}{q}, q \neq 0$$

BODMAS Order

Brackets $\rightarrow$ Orders $\rightarrow \div \times \rightarrow + -$

Irrational Test

Non-terminating, non-repeating decimal

Fractions, decimals and percentages are three representations of the same value. Fluency in converting between them and operating on them is essential for every branch of mathematics.

Fraction Operations

Adding and subtracting fractions requires a common denominator. Multiplying is straightforward; dividing requires flipping the second fraction (using the reciprocal).

► Watch: Algebra – Fractions Review

FRACTION RULES

Add / Subtract

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

Multiply

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

Divide

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

Simplify

Divide numerator & denominator by HCF

1 Mixed fraction operations

Calculate: $\frac{3}{4} + \frac{5}{6} - \frac{1}{3}$

STEP-BY-STEP SOLUTION

- 1 LCM of 4, 6, 3 = 12. Convert: $\frac{3}{4} = \frac{9}{12}$, $\frac{5}{6} = \frac{10}{12}$, $\frac{1}{3} = \frac{4}{12}$
- 2 $\frac{9+10-4}{12} = \frac{15}{12}$
- 3 Simplify: $\text{HCF}(15,12)=3 \rightarrow \frac{15}{12} = \frac{5}{4} = 1\frac{1}{4}$

Answer: $\frac{5}{4} = 1\frac{1}{4}$

A ribbon of length $\frac{7}{2}$ m is cut into pieces each $\frac{7}{8}$ m long. How many pieces are obtained?

STEP-BY-STEP SOLUTION

1 Number of pieces = Total length $\div$ Piece length = $\frac{7}{2} \div \frac{7}{8}$

2 Flip and multiply: $\frac{7}{2} \times \frac{8}{7} = \frac{56}{14} = 4$

Answer: 4 pieces

Percentages & Percentage Change

Percentages express fractions out of 100. Percentage change is used everywhere — discounts, interest rates, growth rates, and statistics.

PERCENTAGE FORMULAS

Percentage of a Value

$$P\% \text{ of } X = \frac{P}{100} \times X$$

% Change

$$\frac{\text{New} - \text{Old}}{\text{Old}} \times 100$$

Fraction $\rightarrow$ %

$$\frac{a}{b} \times 100$$

% $\rightarrow$ Decimal

$$\text{Divide by 100: } 35\% = 0.35$$

1 Discount problem

A laptop costs ₹60,000. It is on sale at 15% off. What is the sale price?

STEP-BY-STEP SOLUTION

1 Discount = $15\% \times 60000 = \frac{15}{100} \times 60000 = 9000$

2 Sale Price = $60000 - 9000 = 51000$

Answer: ₹51,000

2 Percentage increase

A student scored 68 in a test last month and 85 this month. What is the percentage improvement?

STEP-BY-STEP SOLUTION

1 $\% \text{ Change} = \frac{85 - 68}{68} \times 100 = \frac{17}{68} \times 100 = 25\%$

Answer: 25% improvement

3 Reverse percentage

After a 20% increase, the price of a book is ₹360. What was the original price?

STEP-BY-STEP SOLUTION

1 Let original price = P . After 20% increase: $P \times 1.20 = 360$

2 $P = \frac{360}{1.20} = 300$

Answer: ₹300

Module 2 — Formula Summary

Add Fractions

$$\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$$

Multiply Fractions

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

Divide Fractions

$$\frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$$

% Change

$$\frac{\text{New}-\text{Old}}{\text{Old}} \times 100$$

P% of X

$$\frac{P}{100} \times X$$

Ratios compare quantities; proportions state that two ratios are equal. These concepts underpin everything from recipe scaling to financial analysis to physics.

Ratios & Equivalent Ratios

► Watch: Algebra – Ratio & Proportion

RATIO RULES

Ratio

$$a : b = \frac{a}{b}$$

Equivalent Ratio

$$a : b = ka : kb \text{ for any } k \neq 0$$

Divide in Ratio

$$\text{Total} \times \frac{\text{part}}{\text{sum of parts}}$$

Cross Multiplication

$$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$

1 Sharing in a given ratio

Divide ₹2400 among A, B, and C in the ratio 3 : 4 : 5. Find each person's share.

STEP-BY-STEP SOLUTION

- 1 Sum of ratio parts = $3 + 4 + 5 = 12$
- 2 A's share = **Math input error**
- 3 B's share = **Math input error**
- 4 C's share = **Math input error**

Answer: A = ₹600, B = ₹800, C = ₹1000

2 Direct & Inverse Proportion

(a) If 8 workers complete a job in 15 days, how many days will 12 workers take? (b) If a car travels 240 km in 4 hours, how far in 7 hours?

STEP-BY-STEP SOLUTION

- a** More workers $\rightarrow$ fewer days (Inverse): $8 \times 15 = 12 \times d \Rightarrow d = \frac{120}{12} = 10$ days
- b** More time $\rightarrow$ more distance (Direct): $\frac{240}{4} = 60 \text{ km/hr} \rightarrow 7 \times 60 = 420 \text{ km}$

Answer: (a) 10 days; (b) 420 km

Module 3 — Formula Summary

Cross Multiply

$$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$

Direct Proportion

$$y = kx$$

Inverse Proportion

$$xy = k \text{ (constant)}$$

Share formula

$$\text{Total} \times (\text{part} \div \text{sum of parts})$$

Linear equations form the backbone of algebra. Mastering one-variable and two-variable linear equations gives you the toolkit to model almost any real-world situation mathematically.

Solving Linear Equations (One Variable)

► Watch: Linear Equations – Video-Tutor.net

KEY RULES

Standard Form

$$ax + b = c$$

Solution

$$x = \frac{c - b}{a}$$

Golden Rule

Whatever you do to one side, do to the other

Verification

Substitute answer back into original equation

1 Multi-step linear equation

Solve for x : $\frac{2x + 5}{3} - \frac{x - 1}{2} = 4$

STEP-BY-STEP SOLUTION

- 1 Multiply entire equation by LCM(3,2)=6: $2(2x + 5) - 3(x - 1) = 24$
- 2 Expand: $4x + 10 - 3x + 3 = 24$
- 3 Simplify: $x + 13 = 24$
- 4 $x = 11$. Check: $\frac{27}{3} - \frac{10}{2} = 9 - 5 = 4 \checkmark$

Answer: $x = 11$

2 Word problem → linear equation

Ravi is 3 times as old as his son. In 10 years, Ravi will be twice as old as his son. Find their current ages.

STEP-BY-STEP SOLUTION

- 1 Let son's age = x . Then Ravi's age = $3x$.
- 2 In 10 years: $3x + 10 = 2(x + 10)$
- 3 Expand: $3x + 10 = 2x + 20 \rightarrow x = 10$
- 4 Son = 10, Ravi = 30. Check: In 10 years $\rightarrow 20$ and $40 \rightarrow 40 = 2 \times 20 \checkmark$

Answer: Son = 10 years, Ravi = 30 years

Linear Inequalities

► Watch: Inequalities – Video-Tutor.net

INEQUALITY RULES

Addition/Subtraction

Direction unchanged: $\backslash(a$

Multiply/Divide Positive

**Direction unchanged: $\backslash(a0$
 $\rightarrow ac$**

△ Multiply/Divide Negative

Direction FLIPS: $\backslash(abc\backslash$

1 Solve and graph inequality

Solve: $-3x + 7 > 1$ and represent the solution on a number line.

STEP-BY-STEP SOLUTION

- 1 **Subtract 7:**
 $-3x > 1 - 7 = -6$
- 2 **Divide by -3 (FLIP the sign!):** $x < 2$
- 3 **Number line: open circle at 2, shaded to the left ($\leftarrow$)**

**Answer: $x < 2$ i.e.
 $(-\infty, 2)$**

2 Compound inequality

Solve: $-1 \leq 2x - 3 \leq 5$

STEP-BY-STEP SOLUTION

1 Add 3 throughout:

$$-1 + 3 \leq 2x \leq 5 + 3 \rightarrow$$

$$2 \leq 2x \leq 8$$

2 Divide by 2: $1 \leq x \leq 4$

Answer: $x \in [1, 4]$

Module 4 — Formula Summary

Linear Equation

$$ax + b = c \Rightarrow x = \frac{c-b}{a}$$

Inequality: $\div$ by negative

Flip the inequality sign

Solution notation

Closed bracket [] = included, Open

() = excluded

Simultaneous Equations

When two unknowns must be found simultaneously, we use systems of equations. These appear in circuit analysis, economics, geometry, and virtually every applied science.

Substitution Method

► Watch: Systems of Equations – Video-Tutor.net

1 Substitution method

Solve: $y = 2x - 1$ and $3x + y = 14$

STEP-BY-STEP SOLUTION

1 **Substitute** $y = 2x - 1$ into equation 2:

$$3x + (2x - 1) = 14$$

2 $5x - 1 = 14 \Rightarrow 5x = 15 \Rightarrow x = 3$

3 **Back-substitute:** $y = 2(3) - 1 = 5$

Answer: $x = 3, y = 5$

2 Elimination method & word problem

A shop sells pens at ₹5 each and notebooks at ₹15 each. A customer buys 8 items totalling ₹80. How many of each did they buy?

STEP-BY-STEP SOLUTION

1 Let pens = p , notebooks = n . $p + n = 8 \dots$ (i),

$$5p + 15n = 80 \dots$$
 (ii)

- 2 Multiply (i) by 5: $5p + 5n = 40$. Subtract from (ii):
 $10n = 40 \Rightarrow n = 4$

- 3 Substitute: $p + 4 = 8 \Rightarrow p = 4$

Answer: 4 pens and 4 notebooks

Module 5 — Formula Summary

General System

$$a_1x + b_1y = c_1 \text{ and } a_2x + b_2y = c_2$$

Cramer's Rule (x)

$$x = \frac{c_1b_2 - c_2b_1}{a_1b_2 - a_2b_1}$$

Cramer's Rule (y)

$$y = \frac{a_1c_2 - a_2c_1}{a_1b_2 - a_2b_1}$$

Exponents & Radicals (Surds)

Exponent laws are used in every advanced topic — exponential growth, scientific notation, logarithms, calculus power rule. Radicals (surds) are the inverse of exponents.

Laws of Exponents

► Watch: Exponents – Video-Tutor.net

EXPONENT LAWS ($a \neq 0$)

Product

$$a^m \cdot a^n = a^{m+n}$$

Quotient

$$\frac{a^m}{a^n} = a^{m-n}$$

Power of Power

$$(a^m)^n = a^{mn}$$

Zero Exponent

$$a^0 = 1$$

Negative Exponent

$$a^{-n} = \frac{1}{a^n}$$

Fractional Exponent

$$a^{m/n} = \sqrt[n]{a^m}$$

Power of Product

$$(ab)^n = a^n b^n$$

Power of Quotient

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

1 Simplify using exponent laws

Simplify: $\frac{(x^3)^2 \cdot x^{-4}}{x^5}$

STEP-BY-STEP SOLUTION

1 Power of power: $(x^3)^2 = x^6 \rightarrow$ numerator:

$$x^6 \cdot x^{-4} = x^{6+(-4)} = x^2$$

2 **Quotient:** $\frac{x^2}{x^5} = x^{2-5} = x^{-3}$

3 **Rewrite:** $x^{-3} = \frac{1}{x^3}$

Answer: $\frac{1}{x^3}$

2 Simplifying radicals

Simplify: $\sqrt{72} + 2\sqrt{8} - \sqrt{18}$

STEP-BY-STEP SOLUTION

1 $\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$

2 $2\sqrt{8} = 2\sqrt{4 \times 2} = 2 \times 2\sqrt{2} = 4\sqrt{2}$

3 $\sqrt{18} = \sqrt{9 \times 2} = 3\sqrt{2}$

4 **Combine like terms:** $6\sqrt{2} + 4\sqrt{2} - 3\sqrt{2} = 7\sqrt{2}$

Answer: $7\sqrt{2}$

Module 6 — Formula Summary

Product Rule

$$a^m \cdot a^n = a^{m+n}$$

Quotient Rule

$$a^m / a^n = a^{m-n}$$

Power Rule

$$(a^m)^n = a^{mn}$$

Negative Exp

$$a^{-n} = 1/a^n$$

Fractional Exp

$$a^{1/n} = \sqrt[n]{a}$$

Surd Product

$$\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$$

Polynomials & Factoring

Polynomials are the building blocks of algebra. Factoring them is the key skill that unlocks solving higher-degree equations and simplifying rational expressions.

Polynomial Operations & Identities

► Watch: Polynomials – Video-Tutor.net

► Watch: Factoring – Video-Tutor.net

ALGEBRAIC IDENTITIES

$$(a+b)^2 \\ a^2 + 2ab + b^2$$

$$(a-b)^2 \\ a^2 - 2ab + b^2$$

$$a^2 - b^2 \\ (a + b)(a - b)$$

$$a^3 + b^3 \\ (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 \\ (a - b)(a^2 + ab + b^2)$$

$$(a+b)^3 \\ a^3 + 3a^2b + 3ab^2 + b^3$$

1 Factor a trinomial

Factor completely: $6x^2 + 11x - 10$

STEP-BY-STEP SOLUTION

- 1 AC method:** $a = 6, c = -10$, so $ac = -60$. Find two numbers that multiply to -60 and add to 11 : $+15$ and -4 .
- 2 Rewrite middle term:** $6x^2 + 15x - 4x - 10$
- 3 Group:** $3x(2x + 5) - 2(2x + 5) = (3x - 2)(2x + 5)$

Answer: $(3x - 2)(2x + 5)$

2 Difference of cubes

Factor: $8x^3 - 27$

STEP-BY-STEP SOLUTION

- 1 **Identify:** $8x^3 = (2x)^3$ and $27 = 3^3$. This is $a^3 - b^3$ with $a = 2x, b = 3$.
- 2 **Apply identity:** $(a - b)(a^2 + ab + b^2)$
- 3 $= (2x - 3)((2x)^2 + (2x)(3) + 3^2) = (2x - 3)(4x^2 + 6x + 9)$

Answer: $(2x - 3)(4x^2 + 6x + 9)$

Module 7 — Formula Summary

$(a+b)^2$

$$a^2 + 2ab + b^2$$

$(a-b)^2$

$$a^2 - 2ab + b^2$$

Diff. of Squares

$$a^2 - b^2 = (a + b)(a - b)$$

Sum of Cubes

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

Diff. of Cubes

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Quadratic Equations

Quadratics are degree-2 polynomials. Their solutions (roots) can be found by factoring, completing the square, or using the quadratic formula. The discriminant tells you the nature of roots without solving.

Quadratic Formula & Discriminant

► Watch: Quadratic Equations – Video-Tutor.net

QUADRATIC FORMULAS

Standard Form

$$ax^2 + bx + c = 0$$

Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant Δ

$$\Delta = b^2 - 4ac$$

$\Delta > 0$

Two distinct real roots

$\Delta = 0$

One repeated real root

$\Delta < 0$

**Two complex
(imaginary) roots**

Sum of roots

$$\alpha + \beta = -\frac{b}{a}$$

Product of roots

$$\alpha\beta = \frac{c}{a}$$

1 Solve using quadratic formula

Solve: $2x^2 - 5x - 3 = 0$

STEP-BY-STEP SOLUTION

1 $a = 2, b = -5, c = -3.$

$$\Delta = (-5)^2 - 4(2)(-3) = 25 + 24 = 49$$

2 $x = \frac{5 \pm \sqrt{49}}{4} = \frac{5 \pm 7}{4}$

$$3 \quad x_1 = \frac{12}{4} = 3, x_2 = \frac{-2}{4} = -\frac{1}{2}$$

Answer: $x = 3$ or $x = -1/2$

2 Nature of roots & word problem

The length of a rectangle exceeds its width by 5 cm, and its area is 84 cm^2 . Find the dimensions.

STEP-BY-STEP SOLUTION

1 Let width = x . Length = $x + 5$. Area: $x(x + 5) = 84$

2 $x^2 + 5x - 84 = 0$. $\Delta = 25 + 336 = 361 = 19^2 \rightarrow$

Two real roots

3 $x = \frac{-5 \pm 19}{2}$: $x = 7$ or $x = -12$ (reject negative)

4 Width = 7 cm, Length = 12 cm. Check: $7 \times 12 = 84 \checkmark$

Answer: Width = 7 cm, Length = 12 cm

Module 8 — Formula Summary

Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant

$$\Delta = b^2 - 4ac$$

Sum of roots

$$\alpha + \beta = -b/a$$

Product of roots

$$\alpha\beta = c/a$$

$\Delta > 0 \rightarrow 2$ real roots

$\Delta = 0 \rightarrow 1$ real root; $\Delta < 0 \rightarrow$ complex

Geometry & Mensuration

Geometry gives you the tools to calculate areas, perimeters, volumes, and angles. Mensuration (measurement) is widely tested and used in engineering and architecture.

Triangles & Pythagorean Theorem

TRIANGLE FORMULAS

Area

$$\frac{1}{2} \times \text{base} \times \text{height}$$

Heron's Formula

$$A = \sqrt{s(s-a)(s-b)(s-c)}, \quad s = \frac{a+b+c}{2}$$

Pythagoras

$$c^2 = a^2 + b^2 \text{ (right triangle)}$$

Angle Sum

All angles in a triangle sum to 180°

Circle Area

$$\pi r^2$$

Circumference

$$2\pi r$$

Cylinder Volume

$$\pi r^2 h$$

Sphere Volume

$$\frac{4}{3}\pi r^3$$

Cone Volume

$$\frac{1}{3}\pi r^2 h$$

Cube Surface Area

$$6a^2$$

1 Pythagorean theorem

A ladder 13 m long leans against a wall. Its foot is 5 m from the wall. How high does the ladder reach on the wall?

STEP-BY-STEP SOLUTION

1 Hypotenuse = 13, base = 5. $h^2 + 5^2 = 13^2$

2 $h^2 = 169 - 25 = 144 \Rightarrow h = 12 \text{ m}$

Answer: 12 m

2 Heron's formula

Find the area of a triangle with sides 7 cm, 8 cm, 9 cm.

STEP-BY-STEP SOLUTION

1 $s = \frac{7+8+9}{2} = 12$

2 $A = \sqrt{12(12-7)(12-8)(12-9)} = \sqrt{12 \times 5 \times 4 \times 3} =$
 cm^2

Answer: $12\sqrt{5} \approx 26.83 \text{ cm}^2$

3 Volume of a cone

A cone has radius 7 cm and height 12 cm. Find its volume. ($\pi \approx 22/7$)

STEP-BY-STEP SOLUTION

1 $V = \frac{1}{3}\pi r^2 h = \frac{1}{3} \times \frac{22}{7} \times 49 \times 12$

2 $= \frac{1}{3} \times 22 \times 7 \times 12 = \frac{1848}{3} = 616 \text{ cm}^3$

Answer: 616 cm^3

Module 9 — Formula Summary

Pythagoras

$$c^2 = a^2 + b^2$$

Triangle Area

$$\frac{1}{2}bh \text{ or Heron's}$$

Circle Area

$$\pi r^2$$

Circle Perimeter

$$2\pi r$$

Cylinder Vol

$$\pi r^2 h$$

Sphere Vol

$$\frac{4}{3}\pi r^3$$

Cone Vol

$$\frac{1}{3}\pi r^2 h$$

Cube SA

$$6a^2$$

Statistics helps you interpret and summarise data.
Probability quantifies uncertainty. Together they form the language of data science, research, and decision-making.

Measures of Central Tendency & Dispersion

► Watch: Central Tendency – Video-Tutor.net

► Watch: Intro to Statistics – Video-Tutor.net

STATISTICS FORMULAS

Mean

$$\bar{x} = \frac{\sum x_i}{n}$$

Median

Middle value when data is sorted (avg of two middles if n is even)

Mode

Most frequently occurring value

Range

Max – Min

Variance

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

Std Deviation

$$\sigma = \sqrt{\sigma^2}$$

1 Full statistical summary

For the data set: 4, 7, 13, 2, 7, 9, 4, 7 — find mean, median, mode, range, and standard deviation.

STEP-BY-STEP SOLUTION

1 Sorted: 2, 4, 4, 7, 7, 7, 9, 13. $n = 8$

2 Mean: $\bar{x} = \frac{2+4+4+7+7+7+9+13}{8} = \frac{53}{8} = 6.625$

3 Median: Average of 4th & 5th values = $\frac{7+7}{2} = 7$

4 Mode: 7 (appears 3 times)

5 Range: $13 - 2 = 11$

6 Variance: $\sigma^2 = \frac{(2-6.625)^2 + \dots + (13-6.625)^2}{8} \approx 10.23$

7 SD: $\sigma \approx \sqrt{10.23} \approx 3.20$

Mean=6.625, Median=7, Mode=7, Range=11,
SD≈3.20

Basic Probability

► Watch: Introduction to Probability – Video-Tutor.net

PROBABILITY FORMULAS

Basic Probability

$$P(A) = \frac{\text{Favourable outcomes}}{\text{Total outcomes}}$$

Complement

$$P(A') = 1 - P(A)$$

Addition Rule

$$P(A \cup B) = P(A) + P(B)$$

Independent Events

$$P(A \cap B) = P(A) \cdot P(B)$$

Range of P

$$0 \leq P(A) \leq 1$$

1 Deck of cards probability

A card is drawn from a standard 52-card deck. Find: (a) P(Ace), (b) P(Red Card), (c) P(Ace OR Red Card).

STEP-BY-STEP SOLUTION

a $P(\text{Ace}) = \frac{4}{52} = \frac{1}{13}$

b $P(\text{Red}) = \frac{26}{52} = \frac{1}{2}$

c Aces that are Red = 2.

$$P(A \cup R) = \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{28}{52} = \frac{7}{13}$$

$$P(\text{Ace})=1/13, P(\text{Red})=1/2, P(\text{Ace or Red})=7/13$$

2 Independent events

A fair die is rolled and a fair coin is tossed. What is the probability of getting a 5 on the die AND a Head on the coin?

STEP-BY-STEP SOLUTION

1 Events are independent. $P(5) = \frac{1}{6}, P(H) = \frac{1}{2}$

2 $P(5 \text{ AND } H) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$

Answer: 1/12

Module 10 — Formula Summary

Mean

$$\bar{x} = \frac{\sum x_i}{n}$$

Variance

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

Std Dev

$$\sigma = \sqrt{\sigma^2}$$

Probability

$$P = \frac{\text{favourable}}{\text{total}}$$

Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Complement

$$P(A') = 1 - P(A)$$

Master Formula Sheet — School Mathematics

Number & Arithmetic

BODMAS

Brackets → **Orders** → $\div \times$ → $+-$

Rational

$$\frac{p}{q}, q \neq 0$$

% Change

$$\frac{\text{New} - \text{Old}}{\text{Old}} \times 100$$

Cross Multiply

$$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$

Algebra

Linear Solution

$$ax + b = c \Rightarrow x = \frac{c-b}{a}$$

Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum of Roots

$$\alpha + \beta = -b/a$$

Product of Roots

$$\alpha\beta = c/a$$

$(a+b)^2$

$$a^2 + 2ab + b^2$$

$$(a-b)^2$$

$$a^2 - 2ab + b^2$$

$$a^2 - b^2$$

$$(a+b)(a-b)$$

$$a^3 + b^3$$

$$(a+b)(a^2 - ab + b^2)$$

Geometry & Mensuration

Pythagoras

$$c^2 = a^2 + b^2$$

Triangle Area

$$\frac{1}{2}bh$$

Heron's

$$\sqrt{s(s-a)(s-b)(s-c)}$$

Circle Area

$$\pi r^2$$

Circumference

$$2\pi r$$

Cylinder Vol

$$\pi r^2 h$$

Sphere Vol

$$\frac{4}{3}\pi r^3$$

Cone Vol

$$\frac{1}{3}\pi r^2 h$$

Exponents & Radicals

Product

$$a^m \cdot a^n = a^{m+n}$$

Quotient

$$a^m / a^n = a^{m-n}$$

Power

$$(a^m)^n = a^{mn}$$

Negative

$$a^{-n} = 1/a^n$$

Fractional

$$a^{m/n} = \sqrt[n]{a^m}$$

Statistics & Probability

Mean

$$\bar{x} = \frac{\sum x_i}{n}$$

Variance

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

SD

$$\sigma = \sqrt{\sigma^2}$$

Probability

$$P = \frac{\text{favourable}}{\text{total}}$$

Complement

$$P(A') = 1 - P(A)$$

Independent

$$P(A \cap B) = P(A) \cdot P(B)$$

Keep Learning — Watch Free Videos

Explore all topics at [video-tutor.net](https://www.video-tutor.net) — Algebra,
Trigonometry, Calculus, Statistics & more

For college-level mathematics, see Vol. 2: Intermediate College
Mathematics